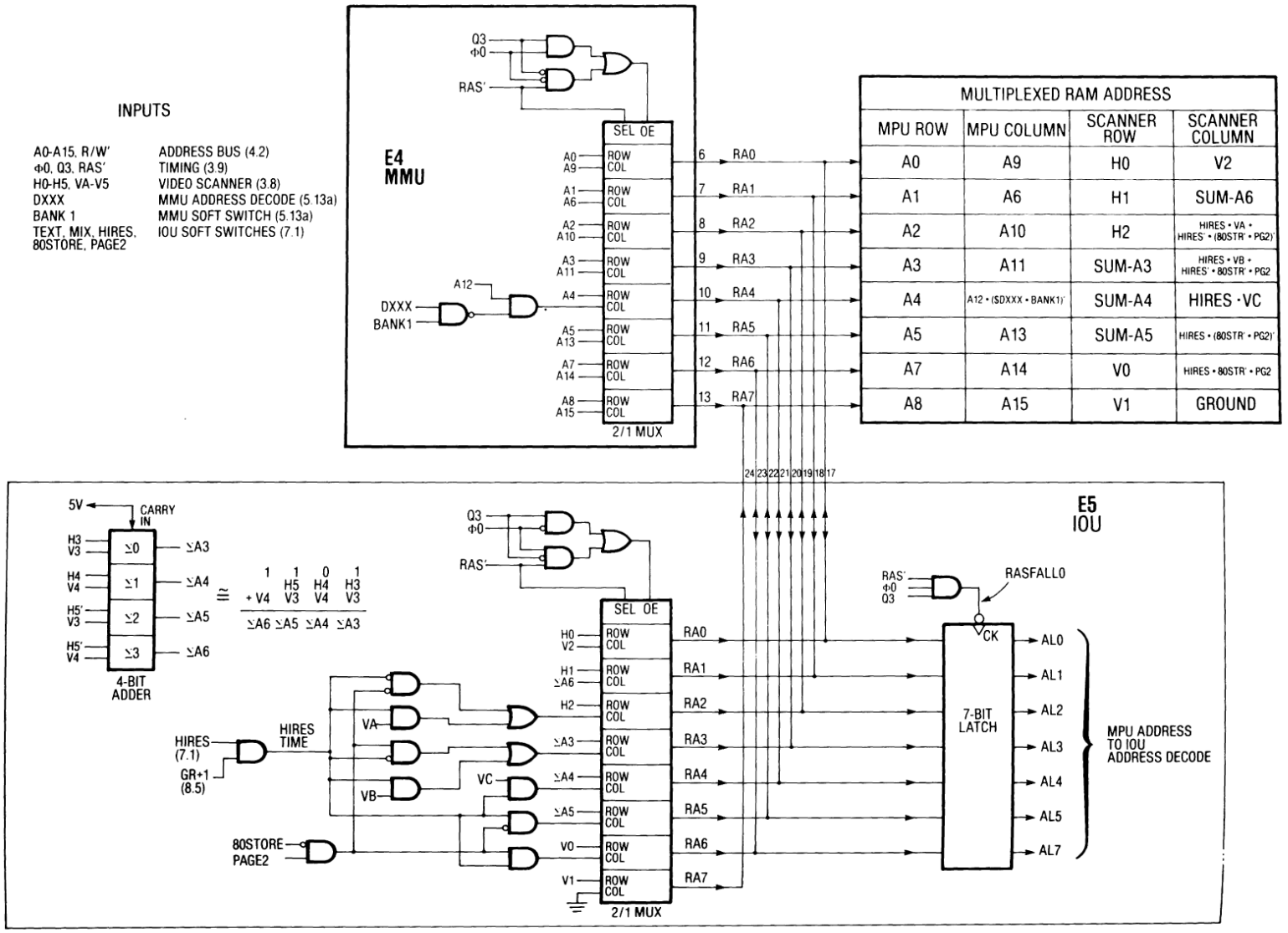


Figure 5.3 Functional Diagram: Apple IIe RAM Address Multiplexing.

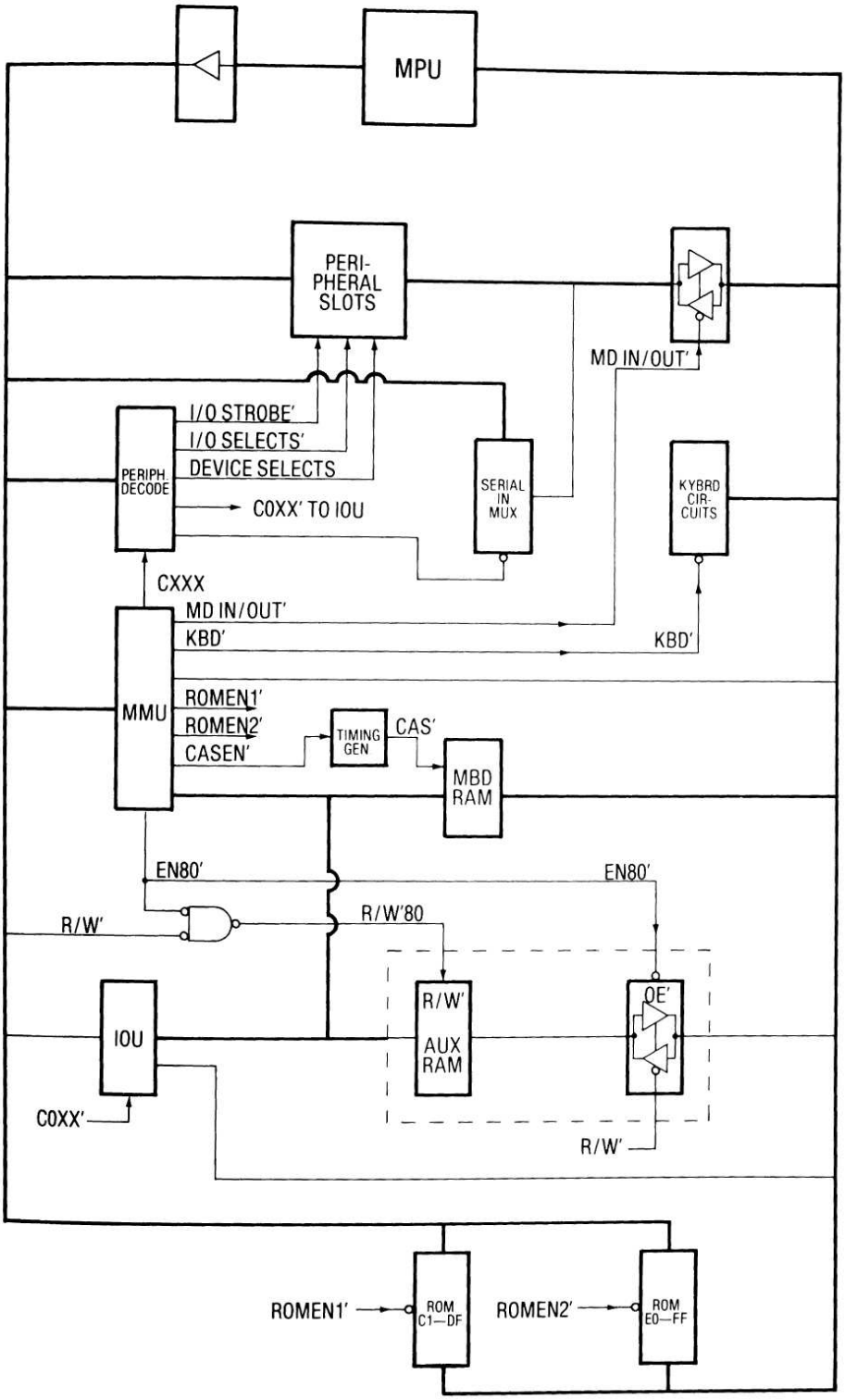


Figure 5.10 Distribution of MMU Data Bus Management Signals.

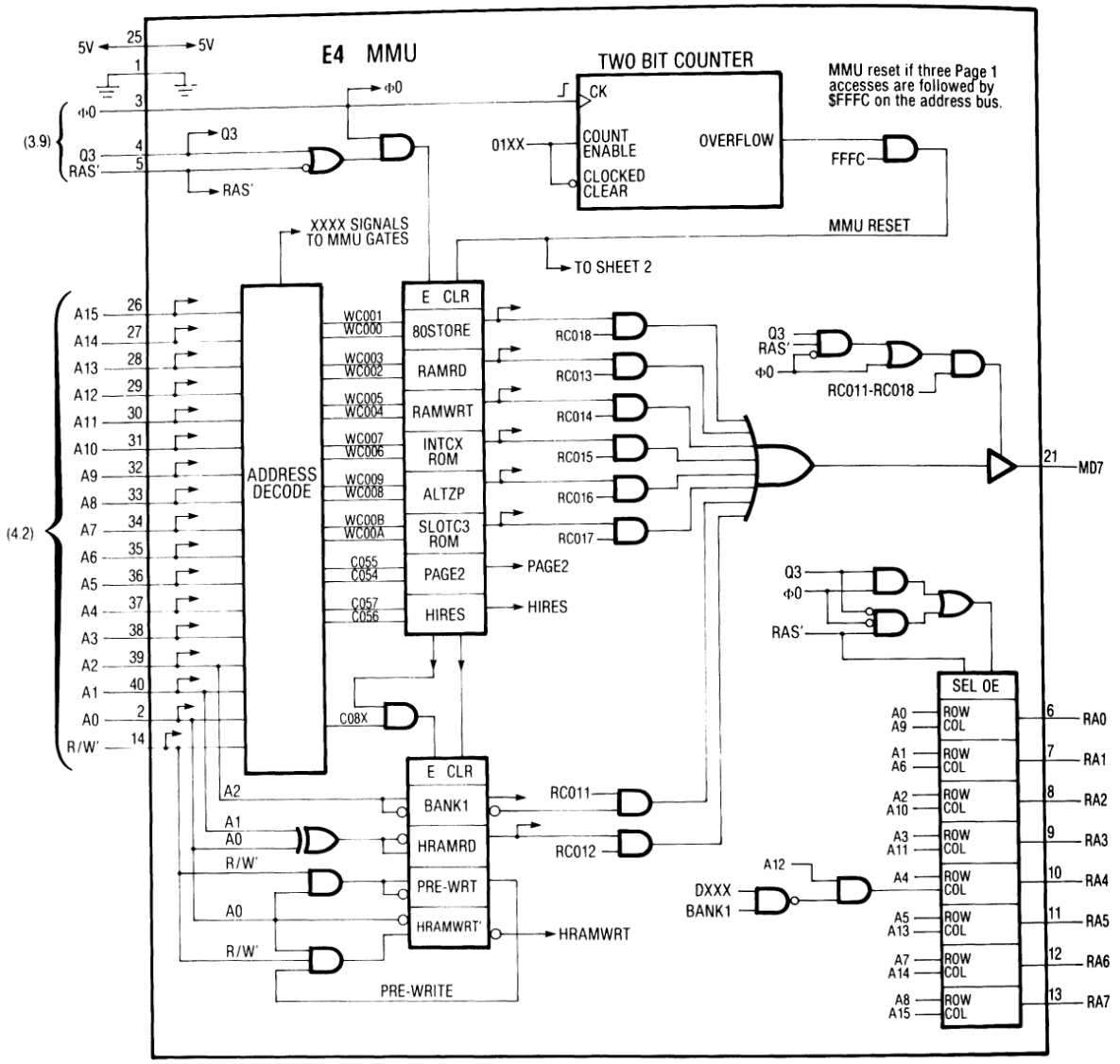
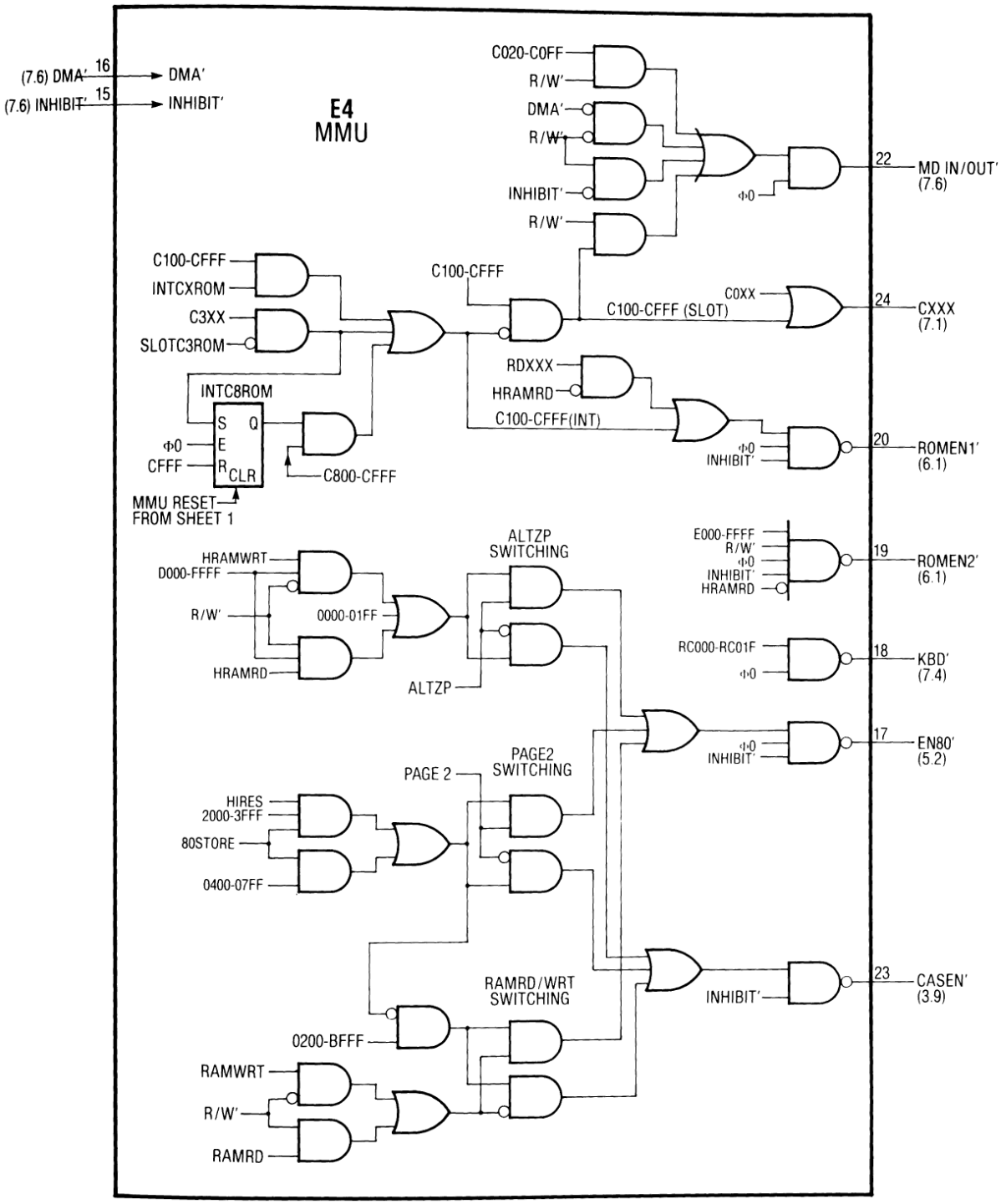


Figure 5.13a The MMU (1 of 2).

6. RAS' selects between ROW and COLUMN addressing at the MPU address multiplexor. The tri-state enable window for the MMU multiplexed RAM address output appears to be PHASE0' • Q3' • RAS' + PHASE0 • Q3 which is the last 14M period of PHASE 1 and the first four 14M periods of PHASE 0.
7. RA4 COLUMN addressing has special logic for switching between bank 1 and bank 2 of high RAM.
8. ROMEN1', ROMEN2', EN80', KBD', and MD IN/OUT' are all gated by PHASE 0. CASEN' is not gated by PHASE 0 in the MMU, but it is gated by PHASE 0 in the timing HAL.
9. All memory enabling signals—ROMEN1', ROMEN2', CASEN', and EN80'—are gated by INHIBIT' from the peripheral slots.
10. CASEN' and EN80' are mutually exclusive. When RAM is being accessed, either CASEN' or EN80', but not both, will be activated.
11. KBD' is gated by RC000—RC01F rather than just the expected RC00X.
12. MD IN/OUT' goes high during any INHIBIT' low read rather than the expected R0000—RBFFF and RC020—RFFFF.
13. Write access to $C100—$CFFF with the addressed location configured for motherboard ROM response causes ROMEN1' to fall. As a



NOTE:
[1] Address range signals decoded from address bus.

Figure 5.13b The MMU (2 of 2).

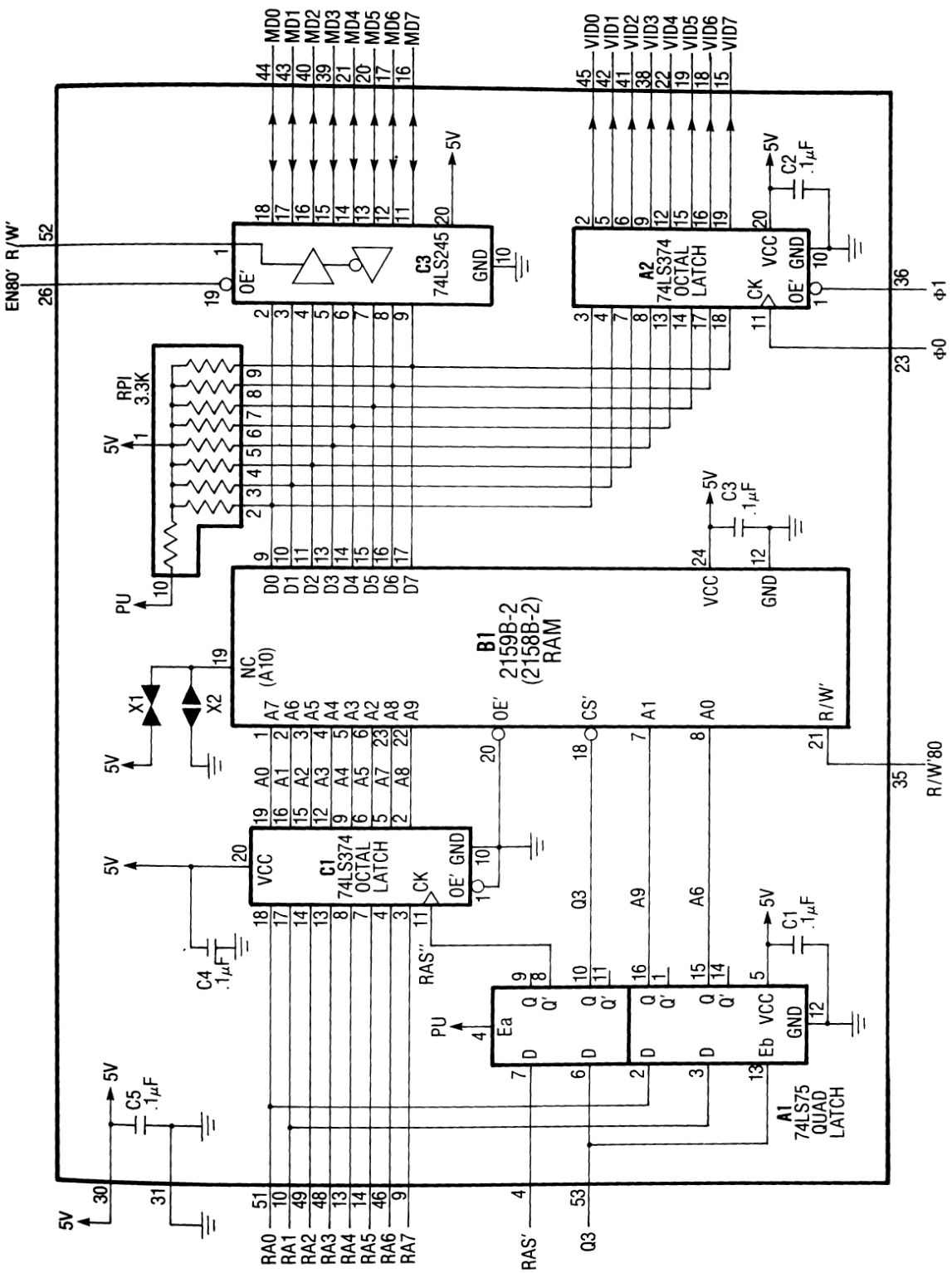
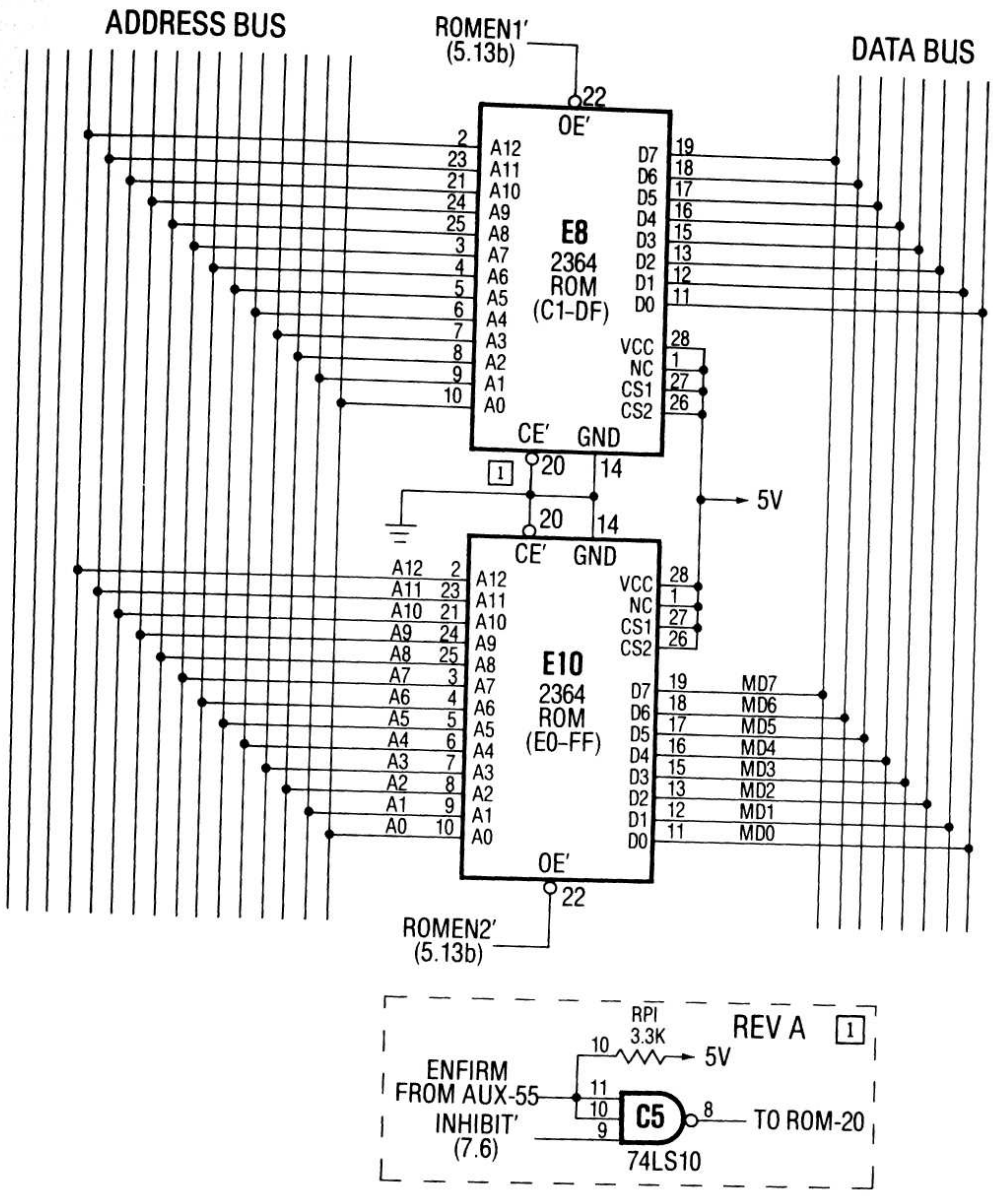


Figure 5.16 Schematic: The 1K Auxiliary RAM Card.



1 In Revision A, ENFIRM and INHIBIT' were required to be high before ROM-20 (CE') would drop low. In Revision B, ENFIRM is replaced by FRCTXT', and INHIBIT' is a logic input to ROMEN1' and ROMEN2' in the MMU.

Figure 6.1 Schematic: ROM in the Apple IIe.

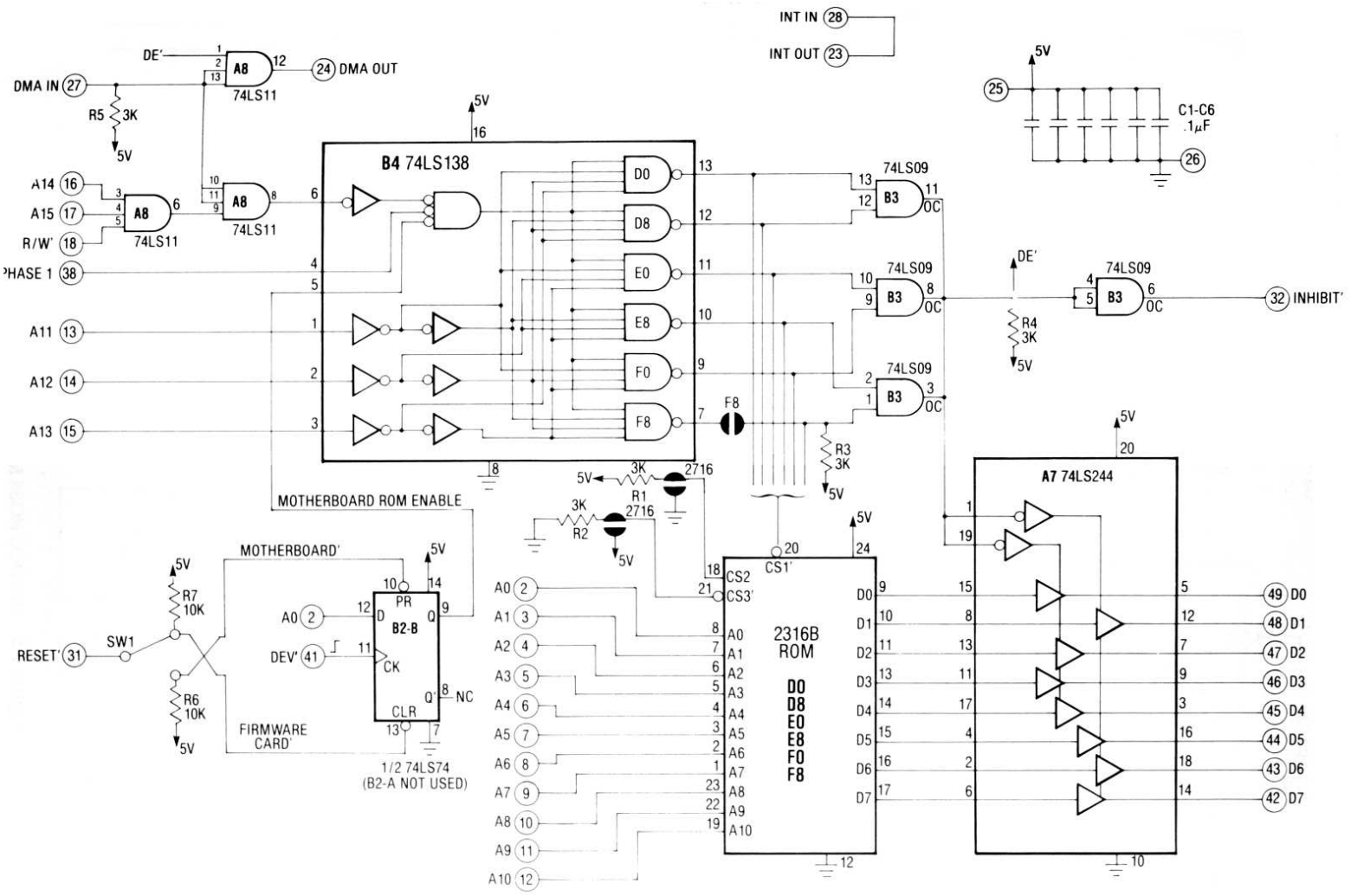


Figure 6.3 Schematic: The 12K Firmware Card.

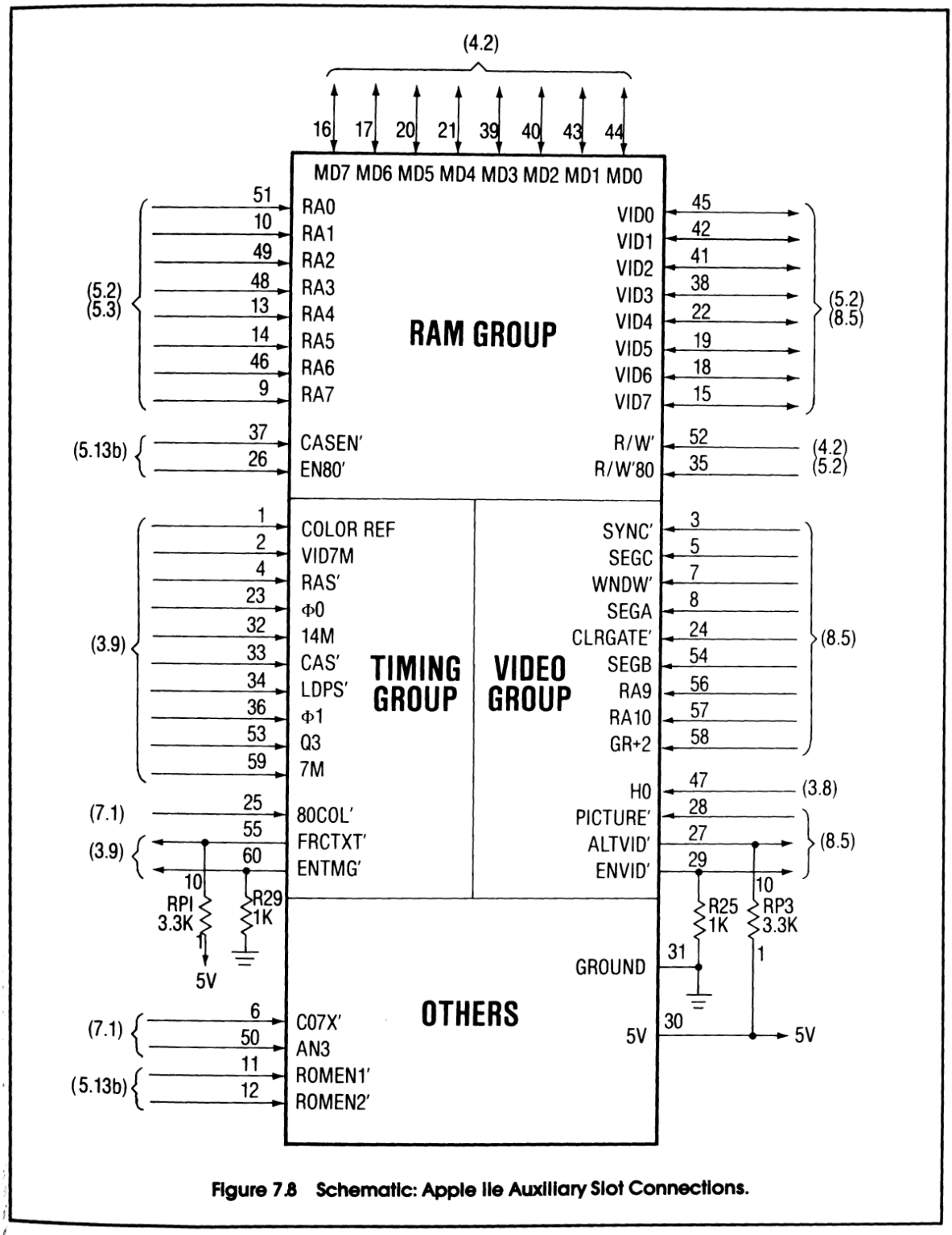


Figure 7.8 Schematic: Apple IIe Auxiliary Slot Connections.